CAREER: Building a Cross-Sector Ecosystem for Trustworthy AI and Cybersecurity Workforce Development

Overview

The rapid advancement of AI, alongside cloud computing, virtual reality, and quantum technologies, is transforming cybersecurity. AI is reshaping how threats are created and defended against, enabling both more sophisticated attacks and intelligent defenses, while increasing the field's complexity and workforce demands. A key challenge is the growing gap between industry needs and workforce readiness, as cybersecurity professionals are increasingly expected to integrate expertise in AI-driven security along with broader cross-disciplinary skills. This rising complexity makes talent development and retention more challenging. Meanwhile, the global workforce gap continues to widen, with an estimated **4.8 million unfilled cybersecurity positions as of 2026**, reflecting significant skill shortages in areas such as cloud security, AI defense, and incident response.

Addressing this challenge requires a shift in education to develop an interdisciplinary and AI-ready cybersecurity workforce. This project proposes a **community-engaged, cross-sector ecosystem** integrating K–12, higher education, and industrial partners to build sustainable pathways in trustworthy AI and cybersecurity. By creating end-to-end "classrooms-to-careers" pathways and embedding industry collaboration, the project aligns training with evolving workforce needs, strengthening the talent pipeline and long-term resilience.

Intellectual Merit

This project introduces an ecosystem-based education framework that integrates "AI for security" with "security of AI" across all levels. Key contributions include: (a) **industry-driven curriculum design** aligned with the NICE framework and evolving workforce needs; (b) an **ecosystem learning model** connecting education, research, and workforce training through K–12, universities, and industry partnerships; (c) **experiential learning** via projects, internships, and industry challenges; and (d) a **scalable, transferable model** to address national workforce needs in AI and cybersecurity.

Broader Impacts

This project will generate impacts across several key areas:

- **Ecosystem Development and Educational Advancement:** Builds a collaborative K–12, university, and industry alliance for AI and cybersecurity education and workforce development. Enhances curricula with interdisciplinary, industry-aligned skills while expanding access to continuous learning and K–12 career pathways.
- **Workforce Empowerment:** Creates multidisciplinary pathways into AI and cybersecurity, strengthening the talent pipeline with job-ready, industry-aligned professionals. Equips the workforce with skills to address evolving challenges, and strengthens the talent pipeline, benefiting employers with a larger, industry-aligned pool of skilled professionals.
- **Economic Growth:** Fuels innovation and growth in AI and cybersecurity by connecting talent with cutting-edge research. Drives business expansion and job creation, while increased economic activity generates tax revenue that supports further regional development.

By integrating education, community engagement, and workforce development, the project advances a resilient and future-ready workforce for secure and trustworthy AI systems.